

Assessment of Density Pattern Retention of Generalized Data for 1:100,000-scale United States Topographic Maps

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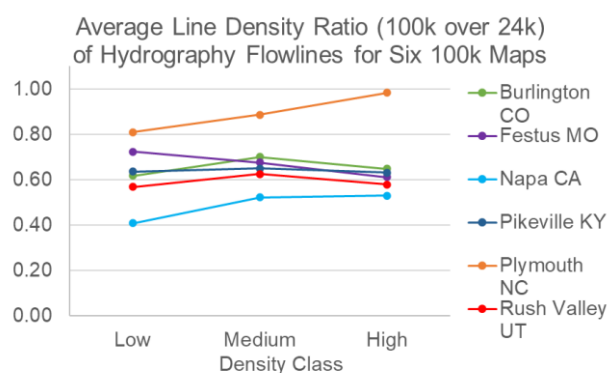
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Abstract:

Cartographic generalization aims to reduce the complexity of geographic data to produce legible smaller-scale displays that retain essential information. Generalization is a vital process in topographic map production. An important challenge in this process is managing and evaluating changes in the density and spatial distribution of map features such as buildings, roads, streams, water bodies, and elevation contours. Density patterns in these features reflect underlying physiographic conditions including factors such as bedrock geology, tectonics, climate and landforms. Assessing acceptable level of change in feature density patterns is critical to ensuring the readability, usability, and accuracy of generalized maps and data.

Methods for assessing the effects of generalization on feature density patterns compare pre- and post-generalization feature densities. Some quantitative approaches for evaluating feature density patterns include kernel density estimation (Gao et al., 2022), grid-based-density metrics (Barrault et al., 2001), entropy-based approaches (Li, Gao, and Xu, 2021), and fractal dimension and scaling behaviour (Goodchild, 1980). The approach presented in this research applies a grid-based density ratio analysis with class-anchored statistical summarization on pre- and post-generalized data. K-means clustering assesses density variation across a map without imposing arbitrary thresholds. The feature density ratio measures the effect of generalization relative to the original content, and the class-based analysis estimates the consistency of the generalization process from low to high-density areas within a map. An example of class-based average density ratios of 1:100,000-scale (100k) over 1:24,000-scale (24k) hydrographic flowlines (e.g., streams, canals, pipelines, artificial flow paths) for six map sheets distributed over the conterminous United States is shown in Figure 1. Map at 100k cover



(a)



(b)

Figure 1. Average density ratios of hydrographic flowline features shown in three density classes for six map sheets (a). The distribution of six map sheets in the conterminous United States (b). Flowline density raster datasets are computed at 1-kilometer resolution for the 1:24,000-scale (24k) and 1:100,000-scale (100k) features, and the average ratio of 100k over 24k densities are computed for each map sheet. A ratio of one indicates there is no change in a class distribution when generalized from 24k to 100k, and lower values indicate part of the 24k feature are retained at 100k.

The purpose of this study is to assess density patterns on the 100k topographic map sheets across the United States. In this work, a generalization workflow that applies a series of operations—such as thinning, simplification, and smoothing—is implemented on the 24k features to generate 100k level of detail data used on 100k topographic maps. To assess density pattern retained in the 100k features across the country, a stratified random sample of at least five percent of the 100k map sheets within each physiographic division of the conterminous United States (Fenneman and Johnson, 1946) are selected for analysis. In addition, at least ten percent of the 100k map sheets within each level II ecological region in Alaska (United States Environmental Protection Agency, 2024) are selected for analysis. The distribution of the selected map sheets is displayed in Figure 2. A 100k map sheet covers about 2100 to 5400 square kilometers, with map sheets getting smaller moving further from the equator.

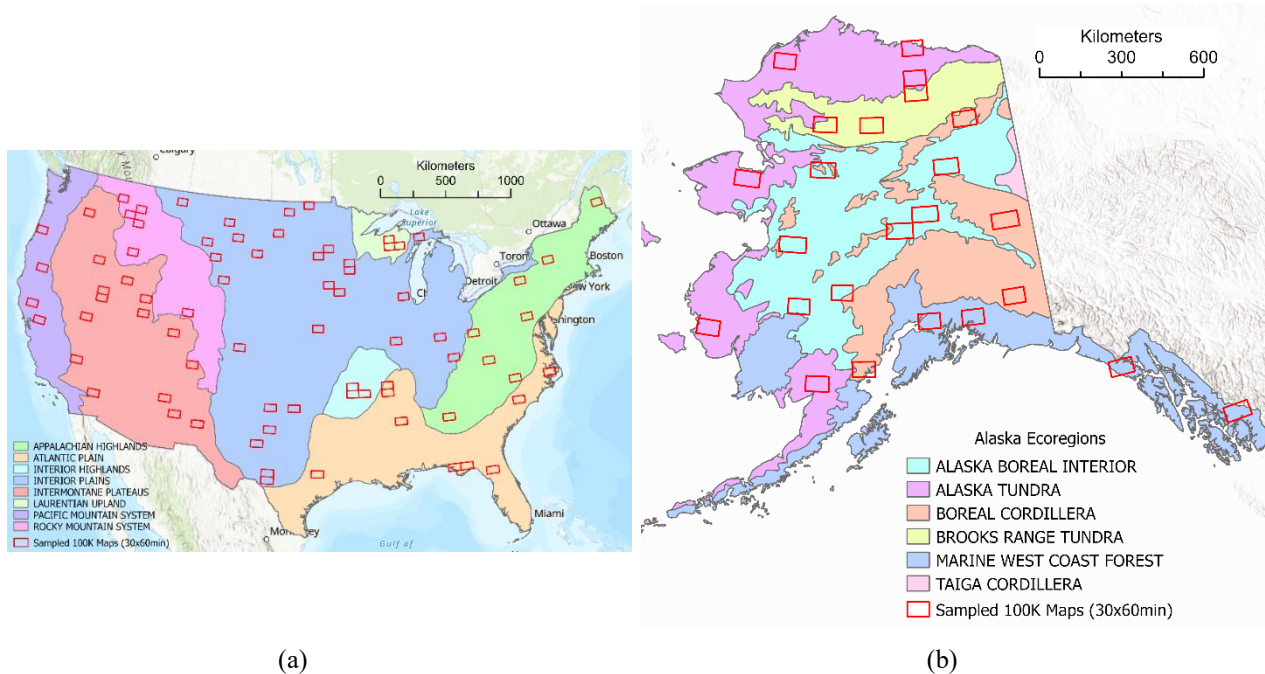


Figure 2. Distribution of stratified random sample of 1:100,000-scale (100k) topographic map sheets in the conterminous United States (a). A five percent random sample of map sheets in each physiographic division (Fenneman and Johnson, 1946) is selected, with at least three map sheets selected per division. Distribution of stratified random sample of 100k topographic map sheets in the Alaska (b). A ten percent random sample of map sheets in each level II ecological region (United States Environmental Protection Agency, 2024) is selected, with at least three map sheets selected per region.

Feature types assessed on each map sheet include hydrography stream channel polylines and other flowlines, polygonal lake, rivers, and other waterbodies, transportation road lines, building point structures, and elevation contour lines. Testing will assess the consistency of feature density pattern retention across individual 100k map sheets and over the different geographic conditions. Results are expected to guide refinements to generalization workflows for topographic mapping at the U.S. Geological Survey.

The software tools described here, and others that are in development to assess the quality of generalized geospatial data, such as positional accuracy and legibility, are being compiled to form an easily accessible set of quality assessment tools for the mapping community, which heretofore have not been readily available. As such tools become more accessible and regularly used in practice, it is expected that these assessment metrics will help form generalization specifications and standards for multiscale topographic maps.

Disclaimer: Any use of trade, firm, or product names is for descriptive purposes only and does not imply endorsement by the U.S. Government.

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